

REM:Rate 12.5—A Versatile Energy Application



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As many *Home Energy* readers know, the Residential Energy Services Network (RESNET) is a U.S.-based, national standards-making nonprofit that defines energy efficiency rating systems. Their best-known system is the HERS index, a numerical value ranging from 0 to over 100 that quantifies the energy efficiency of residential structures. A HERS index of 100 represents the energy used by the American Standard Building and an index of 0 indicates a building that uses no net purchased energy. This rating is used in qualifying homes for energy-efficient mortgages (EEMs), energy improvement mortgages (EIMs), DOE Energy Star Homes and Building America programs, U.S. federal government energy efficiency tax credits, International Energy Conservation Code (IECC) compliance, and most recently, LEED for Homes certification. Clearly, this standard is widely used and well regarded.

In a typical scenario, a RESNET-certified HERS inspector does an on-site inspection of an existing property, measuring its energy characteristics, such as insulation levels, window efficiency, wall-window ratios, heating and cooling system efficiencies, and the solar orientation of the home. Performance testing, such as a blower door test measuring door and duct leakage, may also be used. This information is then entered into a software simulation program that calculates the HERS index. The software may also offer a wide range of additional reports and information, such as

energy upgrades and payback periods, code compliance, and general energy consumption. (Using building plans and specifications, you can calculate a HERS index for a building that has not yet been built.)

There are currently only two RESNET-certified software applications that HERS raters can use to calculate a HERS index for a property. These are REM:Rate Version 12.5, from Architectural Energy Corporation in Boulder, Colorado, and EnergyGauge USA Version 2.7, from Energy Gauge USA in Cocoa, Florida. In this article, I discuss REM:Rate from the perspective of a HERS rater trying to decide if the application is a good choice for his or her situation. I plan to discuss EnergyGauge in a future article.

Getting Started

You can use REM:Rate to model single-family residential homes, multifamily buildings, and mobile homes. Before you start using the program, you need to collect information about the building you are evaluating, either from a site inspection of an existing building or from plans and specifications for a proposed building.

There are two input modes: simplified and detailed. The simplified mode is the easiest way to get started. You first enter data background, such as location, climate zone, and utility company rate schedules. You then enter building characteristics, such as house configuration, conditioned floor area, foundation type, envelope characteristics such as insulation levels and glazing, and mechanical equipment and duct specifications. If you have all the data you need, this process can go quickly, with the possible exception of the glazing information—you need to enter every different combination of window type and orientation.

Unless all your assemblies, such as above-grade walls, ceilings, below-grade walls, floors, and foundations are identical, the simplified mode probably won't

let you describe your building accurately. The detailed mode lets you identify multiple envelope assemblies of the same type (for instance, two floor assemblies with different insulation levels); detailed lighting and appliance schedules; skylights; thermal mass; solar equipment; sunspaces; and much more.

You should be able to enter all the data for a simple building in under an hour. A complex building might take considerably longer, depending on the level of detail you want to track. You can, for instance, enter each door and window individually.

Reporting

Once you complete the data entry, the reporting options are almost overwhelming. Here is a sample of the information you can view and print:

- heating, cooling, hot water, lighting, and appliance energy loads (Btu, MMBtu, or kWh) and dollars;
- energy-efficient mortgage reports;
- Model Energy Code (MEC), ASHRAE, and International Energy Conservation Code (IECC) energy code compliance;
- improvements analysis for existing homes;
- heating and cooling equipment sizing;
- EPA Energy Star Home analysis;
- utility bill reconciliation; and
- HERS and Energy Star certificates.

Collectively, there are more than 60 reports and graphs.

The first thing you'll probably want to do is a Quick Analysis. It summarizes loads and costs; areas (a great way to double-check your data entry skills); and compliance with various standards. From there, you can print a variety of more detailed reports. Some can be customized with your company's logo.

One report that is especially useful for home performance contractors is the Improvements Report. You can enter a set of suggested changes, such as increased insulation levels, improved mechanical equipment, reduced infiltration, and so on. REM:Rate calculates the effect of the improvements on energy consumption, utility costs, and the building's HERS index; recommends improvements based on costs and payback criteria that you enter; and

generates an Energy-Efficient Mortgage Worksheet that can be used to apply for an EEM to finance the upgrades. All in all, it's pretty slick.

Other Features

REM:Rate has a wide range of additional features. For instance, each project has a Notes file and a bare bones spreadsheet that you can use to store additional information about the building. The spreadsheet is a great way to calculate things like building perimeter, total square footage of glazing types on exposures, and wall areas.

Another excellent feature is a set of libraries, for wall and floor assemblies, heating and cooling equipment, improvement measures, and so on, that you can modify. You can either enter new items into a library or add items based on existing library entries. There are also useful functions for managing libraries in a multiuser environment and transferring them between REM:Rate users.

Another feature set that I find interesting is the ability to export a building's data and nearly all calculated output to a MS-Access or SQL Server database. You can then use these data for additional custom reporting or analysis. The product includes templates for creating MS-Access and SQL Server databases in the correct exportable format. The documentation also includes table structures to assist you in customizing the data for your own purposes.

You can also import data from either Elite RHVAC or WrightSoft WrightSuite Residential into REM:Rate. These two products provide ACCA *Manual J*-compatible residential load and duct calculations. REM:Rate imports as many load and building data as it can. You will probably need to do some additional data entry and editing to get the REM:Rate reports you need. I didn't test this feature, or the program's export capabilities.

Documentation

Like most modern software products, REM:Rate does not include a printed manual. Instead, it has a PDF-based manual and an online help system that

borrowed from the PDF manual quite a bit. The content is generally good, with detailed explanations of many concepts and features. Both could be beefed up somewhat with more details on the reports and explanations of some terms. For instance, the description of the Improvements Analysis payback calculation methods only lists the formulas used for the calculations. Some explanation of the pros and cons of each method, along with its applicability, would be helpful for those of us who are not financial wizards.

REM:Rate includes a Getting Started tutorial on the distribution CD-ROM. It's a good quick start guide, but it isn't quite up to date with the actual product. Also it doesn't always synchronize tasks with the viewer instructions. Neither of these is a serious problem. After working through the tutorial, you should be able to start your first project using the simplified input mode.

The installation CD-ROM includes a small number of sample buildings you can study for tips on how to model your own projects. They cover some of the more unusual building types, such as mobile homes and multifamily structures.

Using REM:Rate

REM:Rate is a well-conceived, robust tool for HERS raters and other professionals who need to analyze the energy efficiency and related aspects of new and existing buildings. It's well suited to code compliance and load calculations, and has enough reporting options to accommodate more specialized needs. In addition, Architectural Energy Corporation seems to do a good job of keeping it up to date, to match the latest RESNET standards and requirements.

There are a few things I don't like about REM:Rate. First, it makes use of various scrolling lists of such things as assembly, windows, and types of equipment. With many Windows applications, if you resize a window to make it larger, the number of items in a variable-length list increases to show you more data. Not so with REM:Rate. The number of items stays at five. There's no point in letting you increase

the window size if the additional real estate isn't used.

A second minor complaint is that the error checking is not always clear. For instance, if you enter an invalid day of a month in the utility bill data entry screen, the program substitutes a valid day (such as 2 for 32) without any warning. If you enter an invalid thermostat setpoint temperature (say 45°F), the program automatically changes it to 60°F (its lowest valid value) when you tab out of the field, again without any warning. It makes me nervous to have a piece of software do things behind my back without telling me.

Finally, I just don't like the way REM:Rate organizes reports. The View/Select Reports command displays a dialogue that lets you select individual reports or groups of reports, and you can create your own groups. I just didn't find the dialogue helpful or at all intuitive. After fiddling with it for many minutes, I'm still not really sure how to quickly and easily print out a handful of reports.

Overall, these are minor complaints. HERS providers can license REM:Rate for \$500 per year for internal use or \$1,500 per year if they want to sublicense to other raters. In either case, there is an additional \$6 per HERS rating payable to Architectural Energy Corporation (that fee goes down as a provider's ratings volume goes up). There is also a Research and Oversight license that costs \$500 per year. Intended for researchers, instructors, and HERS QA professionals, this license lets you model buildings but prevents you from doing ratings and Energy Star and tax credit certifications.

For more information:

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To learn more about RESNET, go to www.natresnet.org.

To license REM:Rate, or to learn more about and download Rem:Rate 12.5, go to www.archenergy.com.

To learn more about and download EnergyGauge USA 2.7, go to www.energygauge.com.